

Tech Tales: Digital Storytelling with the Royal BC Museum

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Abstract

In partnership with the Royal BC Museum (RBCM), we are developing an extended reality (XR) experience that breaks the barrier between viewer and museum artifacts through situated engagement. Using 3D models, geospatial data, artifact metadata and curated narratives, the viewer is placed in a virtual space corresponding to the artifacts' stories and guided from artifact to artifact by a conversational AI assistant. Interaction in 3D is facilitated by the use of simple motion detection sensors.

CCS Concepts

• **Information systems** → Digital libraries and archives; • **Applied computing** → Anthropology; Interactive learning environments; • **Human-centered computing** → Natural language interfaces; Virtual reality.

Keywords

Museum and Archives, GIS, Storytelling, VR, XR

ACM Reference Format:

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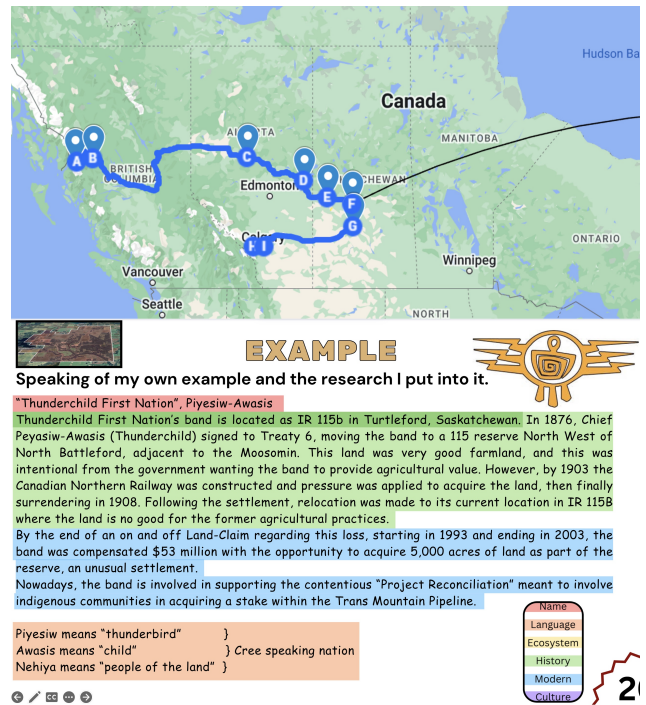


Figure 1: Example of geospatial story telling as a roadtrip between multiple First Nation's Bands (top), with stories using maps, language, history and modern topics (bottom).

Motivation

The Royal BC Museum and Archives makes available troves of visual and written archival materials for the benefit of the public



Figure 2: Screenshots of a VR game developed for the "Gold Rush" and the home screen for a collection, where each artefact floats by within a bubble that can be further selected and interacted with in 3D.

and endurance of recorded knowledge. Photographs depict, audio and videos narrate, and artifacts persist. Together, the observer witnesses a rounded view of the artifact and its story at arm's length. Our project uses virtual reality (VR) to change the way visitors and exhibits co-exist. Through this technology, we plan to support exploration of a distributed set of exhibits. This project will transform access to museum artefacts by bridging collections, empowering visitors to explore vast, interconnected galleries with ease and sometimes even in context with their original environments.

Background Work

These efforts trace back to a group of students who first created an interactive VR game to learn about the gold rush in British Columbia (Figure 2, top). Later, a group of students from the Verna J. Kirkness Program visited Uvic and worked on their vision of *Tech Tales Territories* (Figure 1). This was a collection of stories about where they were from, along with maps and elements of cultural significance. We also built our first application that incorporates motion capture as an interactive technique within virtual environments [Khutorna et al. 2024], and began to collaborate with experts on culturally aware AI [Toxtli et al. 2024].

Key Features and Future Work

Conversations with an AI agent can take place in any of 57 supported languages, or even a combination of languages! The semantic understanding of the objects and their history is reliable and accurate, through extensive LLM evaluation in addition to observation from domain experts. Exploration of artefacts can be situated in virtual geographic settings, and interaction can be through simple motion detection.

The AI assistant (a seal, Figure 2, bottom) shows people a selection of objects and experiences from a set of museum collections and virtual tours, and the viewer can begin their exploration from any one of these "bubbles". One such virtual tour is a Salish Sea dive into the world of single cellular Diatoms, otherwise invisible to the naked eye. Voice interaction with the seal is ongoing in a conversational manner, and based on their level of engagement with different aspects of the object. For example, if they view a Diatom, and they only ask questions or are more engaged with a Salish Sea theme, they would be forwarded to another object from this microcosm of organisms. Conversely, if they ask questions about art, they would be directed to more art, of a style based on the

questions they ask. As geospatial context arises in the AI response pertaining to an object, are transported to the relevant location.

References

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